

Gauge Algebra and Hypercharge Structure from the Hodge Complex on the 8×8 Torus

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September 23, 2026

Abstract

We prove nine theorems about the Hodge complex of differential forms on the periodic lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.

(T1) The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ on each form degree $k = 0, 1, 2$, where Δ is the discrete 5-point Laplacian.

(T2) The eigenvalue $\lambda = 4$ of Δ has multiplicity 14 on Ω^0 , realized on the fourteen lattice momenta $\mathcal{M}$ given explicitly in equation (14).

(T3) The D_4 representation decompositions at $\lambda = 4$ are

$$\begin{aligned}\Omega^0|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \\ \Omega^1|_{\lambda=4} &= 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \\ \Omega^2|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E.\end{aligned}$$

(T4) The D_4 commutants have dimensions 27 on $\Omega^0|_{\lambda=4}$ and 100 on $\Omega^1|_{\lambda=4}$, computed from Schur’s lemma.

(T5) The algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ embeds in the commutant on $\Omega^0|_{\lambda=4}$ as a direct sum of Lie subalgebras.

(T6) The Hodge–Dirac evades Nielsen–Ninomiya: the zero-mode dimensions are the Betti numbers $(b_0, b_1, b_2) = (1, 2, 1)$, and the nonzero spectrum is $\pm$ -symmetric.

(T7) The nine A_1 copies in the $\lambda = 4$ sector split 3/3/3 across the three Hodge sectors.

(T8) The space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators on the 56-dimensional $\lambda = 4$ sector is 15-dimensional.

(T9) The Hodge complex admits an orientation-flip $\mathbb{Z}_2$ symmetry that swaps $\Omega^0 \leftrightarrow \Omega^2$ while fixing Ω^1 . Under this symmetry, the physical content of the framework is invariant up to the relabeling $u_R^c \leftrightarrow d_R^c$.

The paper’s two inputs are stated explicitly in Section 1.2: the Standard Model per-generation content, and the physical identification of the A_1 isotypic with color and the B_1 isotypic with weak isospin. Theorems T1–T9 depend only on the Hodge complex and these two identifications.

All proofs are analytic or by finite computation on a 256-dimensional real vector space.

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1 Introduction and inputs

1.1 The Hodge complex

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the periodic square lattice with $N = 64$ sites. The Hodge complex of differential forms on Λ is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}. \quad (1)$$

The exterior derivative d and codifferential d^* are the standard lattice operators. The Hodge–Dirac operator is $D = d + d^*$, a Hermitian operator on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ of total dimension 256.

1.2 The two inputs

The paper has two inputs, stated here and used explicitly throughout.

Input 1 (Standard Model content). The Standard Model per-generation content, in standard left-handed convention:

$$\begin{aligned} Q_L(3, 2, +\tfrac{1}{6}) \times 6, & \quad u_R^c(3, 1, -\tfrac{2}{3}) \times 3, & \quad d_R^c(3, 1, +\tfrac{1}{3}) \times 3, \\ L_L(1, 2, -\tfrac{1}{2}) \times 2, & \quad e_R^c(1, 1, +1) \times 1, & \quad \nu_R^c(1, 1, 0) \times 1. \end{aligned} \quad (2)$$

Input 2 (Physical identification). The identification of the A_1 isotypic of the D_4 commutant with the physical color algebra, and of the B_1 isotypic with the physical weak isospin algebra.

The framework derives the algebraic structure $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ from the Hodge complex. The identification of this algebraic structure with the physical color and weak-isospin gauge groups is Input 2. This is the same physical identification the Standard Model itself makes: the Standard Model takes its gauge group as empirical input, and the WIN Paradigm derives a unique candidate for the algebraic structure that must carry the physical gauge algebra.

1.3 What is not claimed

We do **not** claim to derive the specific hypercharge values $-2/3, +1/6, +1/3, -1/2, +1, 0$. The framework derives the 15-dimensional space in which any hypercharge functional must live. The Standard Model content, taken as Input 1, selects one point in that space. The paper computes the unique coefficients reproducing the Standard Model values and proves that the resulting configuration is anomaly-free.

We do **not** claim to derive the Standard Model content. The framework derives the structure that this content must satisfy.

We do **not** privilege the sector Ω^0 over Ω^2 , or vice versa. The two sectors are Hodge-dual to each other and carry identical D_4 decompositions. Theorem 9.1 (below) establishes the orientation-flip $\mathbb{Z}_2$ symmetry that exchanges them, and shows that all physical predictions of the framework are invariant under this exchange.

1.4 Organization

Section 2 constructs the Hodge complex and proves the square identity. Section 3 derives the $\lambda = 4$ decomposition. Section 4 computes the commutant dimensions. Section 5 constructs the gauge algebra embedding. Section 6 proves Nielsen–Ninomiya evasion. Section 7 proves the sector-triplet correspondence. Section 8 derives the 15-dimensional hypercharge space. Section 9 proves the orientation-flip symmetry. Section 10 verifies anomaly cancellation as a consistency condition on Input 1. Section 11 summarizes what is derived and what is input.

2 The Hodge complex

2.1 Lattice and forms

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Let $\{e_1, e_2\}$ be the standard basis. Sites are denoted $x = (x_1, x_2) \in \Lambda$. The space of k -forms is:

Ω^0 : real functions on sites;

Ω^1 : real 1-forms ω with components $\omega(x, \mu)$ for $\mu \in \{1, 2\}$ and $x \in \Lambda$;

Ω^2 : real 2-forms χ with one component per site.

2.2 The exterior derivative

The exterior derivative is defined by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad (3)$$

$$(d_1 \omega)(x) = \omega(x, 1) + \omega(x + e_1, 2) - \omega(x + e_2, 1) - \omega(x, 2). \quad (4)$$

Lemma 2.1. $d_1 d_0 = 0$.

Proof. Direct computation:

$$\begin{aligned}
(d_1 d_0 f)(x) &= (d_0 f)(x, 1) + (d_0 f)(x + e_1, 2) - (d_0 f)(x + e_2, 1) - (d_0 f)(x, 2) \\
&= [f(x + e_1) - f(x)] + [f(x + e_1 + e_2) - f(x + e_1)] \\
&\quad - [f(x + e_2 + e_1) - f(x + e_2)] - [f(x + e_2) - f(x)] \\
&= 0. \quad \square
\end{aligned}$$

□

2.3 The codifferential

The codifferential is defined by the adjoint property with respect to the standard inner products

$$\langle f, g \rangle_{\Omega^0} = \sum_{x \in \Lambda} f(x)g(x), \quad \langle \omega, \eta \rangle_{\Omega^1} = \sum_{x, \mu} \omega(x, \mu)\eta(x, \mu). \quad (5)$$

Explicitly,

$$(d_0^* \omega)(x) = \sum_{\mu=1}^2 [\omega(x - e_\mu, \mu) - \omega(x, \mu)], \quad (6)$$

$$(d_1^* \chi)(x, \mu) = \chi(x) - \chi(x + e_\mu). \quad (7)$$

Lemma 2.2. $\langle d_0 f, \omega \rangle_{\Omega^1} = \langle f, d_0^* \omega \rangle_{\Omega^0}$.

Proof. Direct computation:

$$\langle d_0 f, \omega \rangle_{\Omega^1} = \sum_{x, \mu} [f(x + e_\mu) - f(x)]\omega(x, \mu) = \sum_{x, \mu} f(x)[\omega(x - e_\mu, \mu) - \omega(x, \mu)] = \langle f, d_0^* \omega \rangle_{\Omega^0}. \quad \square$$

□

2.4 The Hodge–Dirac operator

Define D on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ by

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}. \quad (8)$$

Theorem 2.3 (Hodge–Dirac square). *For $k = 0, 1, 2$,*

$$D^2|_{\Omega^k} = \Delta, \quad (9)$$

where Δ is the discrete 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{\mu=1}^2 [f(x + e_\mu) + f(x - e_\mu)]. \quad (10)$$

Proof. First consider $k = 0$. From (8), $(D^2 f)|_{\Omega^0} = d_0^* d_0 f$ for $f \in \Omega^0$. Direct computation:

$$\begin{aligned}
(d_0^* d_0 f)(x) &= \sum_{\mu} [(d_0 f)(x - e_\mu, \mu) - (d_0 f)(x, \mu)] \\
&= \sum_{\mu} [(f(x) - f(x - e_\mu)) - (f(x + e_\mu) - f(x))] \\
&= \sum_{\mu} [2f(x) - f(x + e_\mu) - f(x - e_\mu)] \\
&= 4f(x) - \sum_{\mu=1}^2 [f(x + e_\mu) + f(x - e_\mu)] = (\Delta f)(x).
\end{aligned}$$

The cases $k = 1, 2$ follow by identical computation using (7). $\square$

Corollary 2.4. $[D^2, P_{\Omega^k}] = 0$ for each sector projector P_{Ω^k} .

2.5 The D_4 action

The dihedral group $D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle$ acts on Λ by the point group of the square:

$$r : (x_1, x_2) \mapsto (-x_2, x_1), \quad s : (x_1, x_2) \mapsto (x_1, -x_2), \quad (11)$$

modulo 8. On Ω^0 and Ω^2 , D_4 permutes sites. On Ω^1 , D_4 acts with the vector twist:

$$r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1), \quad s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2). \quad (12)$$

Proposition 2.5. $[R, D^2] = [S, D^2] = 0$ and $[R, P_{\Omega^k}] = [S, P_{\Omega^k}] = 0$.

Proof. The point-group action commutes with the Laplacian by translation invariance of Δ . The vector twist on Ω^1 is the standard representation of D_4 on $\mathbb{R}^2$. Direct check. $\square$

3 The $\lambda = 4$ eigenvalue

3.1 Spectrum of the Laplacian

The eigenfunctions of Δ on Λ are plane waves $\phi_{n_1, n_2}(x) = e^{2\pi i(n_1 x_1 + n_2 x_2)/8}$ for $(n_1, n_2) \in \{0, \dots, 7\}^2$, with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (13)$$

Lemma 3.1. The spectrum of Δ is mirror-symmetric: $\lambda(n_1, n_2) = \lambda(8 - n_1, 8 - n_2)$.

Proof. $\cos(\pi(8 - n)/4) = \cos(2\pi - \pi n/4) = \cos(\pi n/4)$. $\square$

3.2 Multiplicity of $\lambda = 4$

Theorem 3.2. The eigenvalue $\lambda = 4$ has multiplicity 14, realized on the set

$$\begin{aligned} \mathcal{M} = & \{(0, 4), (4, 0)\} \\ & \cup \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\} \\ & \cup \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (14)$$

Proof. Setting $\lambda(n_1, n_2) = 4$ gives

$$\cos\left(\frac{\pi n_1}{4}\right) + \cos\left(\frac{\pi n_2}{4}\right) = 0. \quad (15)$$

The values of $\cos(\pi n/4)$ for $n = 0, \dots, 7$ are $1, \sqrt{2}/2, 0, -\sqrt{2}/2, -1, -\sqrt{2}/2, 0, \sqrt{2}/2$. Equation (15) requires the two cosine values to be negatives of each other. The allowed pairs are:

$(1, -1)$: $n_1 = 0, n_2 = 4$. One solution.

$(-1, 1)$: $n_1 = 4, n_2 = 0$. One solution.

$(\sqrt{2}/2, -\sqrt{2}/2)$: $n_1 \in \{1, 7\}, n_2 \in \{3, 5\}$. Four solutions.

$(-\sqrt{2}/2, \sqrt{2}/2)$: $n_1 \in \{3, 5\}, n_2 \in \{1, 7\}$. Four solutions.

$(0, 0)$: $n_1 \in \{2, 6\}, n_2 \in \{2, 6\}$. Four solutions.

Total: $1 + 1 + 4 + 4 + 4 = 14$. $\square$

3.3 The D_4 orbit decomposition

Lemma 3.3. *Under D_4 , the set $\mathcal{M}$ partitions into three orbits:*

$$\mathcal{O}_1 = \{(0, 4), (4, 0)\}, \quad |\mathcal{O}_1| = 2, \quad (16)$$

$$\mathcal{O}_2 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad |\mathcal{O}_2| = 8, \quad (17)$$

$$\mathcal{O}_3 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad |\mathcal{O}_3| = 4. \quad (18)$$

Proof. Direct orbit computation. The orbit of $(0, 4)$: $s(0, 4) = (0, -4) = (0, 4)$; $r(0, 4) = (-4, 0) = (4, 0)$; $r(4, 0) = (0, 4)$. Orbit $\{(0, 4), (4, 0)\}$, size 2.

The orbit of $(1, 3)$: applying r, s generates $(1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)$. Orbit size 8.

The orbit of $(2, 2)$: $r(2, 2) = (-2, 2) = (6, 2)$; $s(2, 2) = (2, 6)$; $r(2, 6) = (-6, 2) = (2, 2)$; orbit $\{(2, 2), (2, 6), (6, 2), (6, 6)\}$, size 4. $\square$

3.4 The D_4 representation decomposition

Character table of D_4 . With conjugacy classes $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$:

irrep	e	r^2	r	s	rs
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

Character of the 14-mode space. Let $V = \Omega^0|_{\lambda=4}$ with basis $\{\phi_m : m \in \mathcal{M}\}$. The D_4 action permutes the modes. A mode m contributes to $\chi_V(g)$ if and only if $g \cdot m = m$.

$$\chi_V(e) = 14.$$

$\chi_V(r) = 0$: fixed modes satisfy $(-n_2, n_1) = (n_1, n_2) \pmod{8}$, giving $(0, 0), (4, 4)$, neither in $\mathcal{M}$.

$$\chi_V(r^2) = 2$$
: fixed modes satisfy $(0, 4), (4, 0)$.

$$\chi_V(s) = 2$$
: fixed modes $(0, 4), (4, 0)$.

$$\chi_V(rs) = 2$$
: fixed modes $(2, 2), (6, 6)$.

Theorem 3.4 (Decomposition of $\Omega^0|_{\lambda=4}$).

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (19)$$

Proof. Multiplicities

$$m_\rho = \frac{1}{8} \sum_{\text{classes } C} |C| \chi_V(C) \chi_\rho(C). \quad (20)$$

Computing:

$$m_{A_1} = \frac{1}{8} [1 \cdot 14 \cdot 1 + 1 \cdot 2 \cdot 1 + 2 \cdot 0 \cdot 1 + 2 \cdot 2 \cdot 1 + 2 \cdot 2 \cdot 1] = 3,$$

$$m_{A_2} = \frac{1}{8} [14 - 0 - 4 - 4] = 1,$$

$$m_{B_1} = \frac{1}{8} [14 - 0 + 4 - 4] = 2,$$

$$m_{B_2} = \frac{1}{8} [14 - 0 - 4 + 4] = 2,$$

$$m_E = \frac{1}{8} [2 \cdot 14 + 1 \cdot 2 \cdot (-2)] = 3.$$

Dimension check: $3 + 1 + 2 + 2 + 6 = 14$. $\square$

Corollary 3.5 (Decomposition of $\Omega^1|_{\lambda=4}$). *The two vector components of Ω^1 transform as the E irrep. Using $A_i \otimes E = B_i \otimes E = E$ and $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$:*

$$\Omega^1|_{\lambda=4} = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E. \quad (21)$$

Dimension check: $3 + 3 + 3 + 3 + 16 = 28$.

Corollary 3.6 (Decomposition of $\Omega^2|_{\lambda=4}$). *By Hodge duality,*

$$\Omega^2|_{\lambda=4} = \Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (22)$$

Corollary 3.7 (Total content).

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (23)$$

Dimension check: $9 + 5 + 7 + 7 + 28 = 56$.

4 The D_4 commutant

Definition 4.1. *The D_4 commutant on a D_4 -representation V is*

$$\text{Comm}(D_4, V) = \{X \in \text{End}(V) : [X, R] = [X, S] = 0\}. \quad (24)$$

Theorem 4.2 (Schur). *If $V = \bigoplus_{\rho} m_{\rho} V_{\rho}$ where the V_{ρ} are irreducible, then*

$$\dim \text{Comm}(D_4, V) = \sum_{\rho} m_{\rho}^2. \quad (25)$$

Proof. The algebra of D_4 -intertwiners on V is isomorphic to $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$, with dimension $\sum_{\rho} m_{\rho}^2$. $\square$

Corollary 4.3. $\dim \text{Comm}(D_4, \Omega^0|_{\lambda=4}) = 3^2 + 1^2 + 2^2 + 2^2 + 3^2 = 27$.

Corollary 4.4. $\dim \text{Comm}(D_4, \Omega^1|_{\lambda=4}) = 3^2 + 3^2 + 3^2 + 3^2 + 8^2 = 100$.

Corollary 4.5. $\dim \text{Comm}(D_4, \text{full } 56\text{-dim}) = 9^2 + 5^2 + 7^2 + 7^2 + 14^2 = 400$.

5 Gauge algebra embedding

5.1 The $\mathfrak{su}(3)$ on the A_1 multiplicity space

The A_1 isotypic of $\Omega^0|_{\lambda=4}$ has multiplicity 3. Let $\{v_1, v_2, v_3\}$ be an orthonormal basis. Define

$$\phi : \mathfrak{gl}(3, \mathbb{R}) \rightarrow \text{End}(\Omega^0|_{\lambda=4}), \quad \phi(M) = \sum_{a,b=1}^3 v_a M_{ab} v_b^T. \quad (26)$$

Proposition 5.1. *ϕ is a Lie algebra homomorphism: $\phi([M, N]) = [\phi(M), \phi(N)]$.*

Proof. Direct computation:

$$[\phi(M), \phi(N)] = \sum_{a,b,c,d} v_a M_{ab} v_b^T v_c N_{cd} v_d^T - (M \leftrightarrow N) = \sum_{a,d} v_a (MN - NM)_{ad} v_d^T = \phi([M, N]). \quad \square$$

$\square$

Proposition 5.2. $\phi(\mathfrak{gl}(3, \mathbb{R})) \subseteq \text{Comm}(D_4, \Omega^0|_{\lambda=4})$.

Proof. The A_1 isotypic is a single D_4 -irrep; on it, D_4 acts trivially. Therefore any operator preserving the isotypic decomposition commutes with D_4 . $\square$

Definition 5.3. The eight generators of $\mathfrak{su}(3)_c$ are $T_c^a = \phi(\lambda^a)$ for the Gell-Mann matrices λ^a , $a = 1, \dots, 8$:

$$\begin{aligned} \lambda^1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda^4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda^5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & \lambda^6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ \lambda^7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda^8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \end{aligned} \quad (27)$$

The generators satisfy $[T_c^a, T_c^b] = i \sum_c f^{abc} T_c^c$ with f^{abc} the $\mathfrak{su}(3)$ structure constants.

5.2 The $\mathfrak{su}(2)$ on the B_1 multiplicity space

The B_1 isotypic of $\Omega^0|_{\lambda=4}$ has multiplicity 2. Let $\{w_1, w_2\}$ be an orthonormal basis. Define

$$T_L^i = \sum_{j,k=1}^2 w_j \sigma_{jk}^i w_k^T, \quad i = 1, 2, 3, \quad (28)$$

with σ^i the Pauli matrices.

Proposition 5.4. $[T_L^i, T_L^j] = i \sum_k \epsilon^{ijk} T_L^k$.

Proof. Same as above with $\mathfrak{gl}(2, \mathbb{R}) \cong \mathfrak{u}(2)$. $\square$

5.3 The $\mathfrak{u}(1)$ factors

The five isotypic projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$ span an abelian subalgebra of $\text{Comm}(D_4, \Omega^0|_{\lambda=4})$, isomorphic to $\mathfrak{u}(1)^5$.

5.4 Closure

Theorem 5.5. The twelve generators T_c^a (eight), T_L^i (three), and P_{A_1} (one) generate a Lie subalgebra of $\text{Comm}(D_4, \Omega^0|_{\lambda=4})$ isomorphic to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

Proof. (1) $[T_c^a, T_c^b] = i f^{abc} T_c^c$.

$$(2) [T_L^i, T_L^j] = i \epsilon^{ijk} T_L^k.$$

$$(3) [T_c^a, T_L^i] = 0: T_c^a \text{ acts on } A_1, T_L^i \text{ acts on } B_1; \text{ the isotypics are orthogonal.}$$

$$(4) [T_c^a, P_{A_1}] = 0: \text{ both act on } A_1; P_{A_1} \text{ is the identity there. Similarly for } T_L^i \text{ with } P_{A_1}.$$

$$(5) P_{A_1} \text{ is abelian.}$$

The generators close to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. $\square$

6 Nielsen–Ninomiya evasion

Theorem 6.1. *The Hodge–Dirac D on the discrete torus $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$, $L \geq 2$:*

- (i) *is local (nearest-neighbour hopping), Hermitian, and translation-invariant;*
- (ii) *has $\dim(\ker D \cap \Omega^k) = b_k(T^2)$ for $k = 0, 1, 2$;*
- (iii) *has a nonzero spectrum that is exactly $\pm$ -symmetric.*

Proof. (i) D is the sum of nearest-neighbour hoppings defined in (3)–(7). Hermiticity follows from Lemma 2.2. Translation-invariance is manifest.

(ii) By the Hodge isomorphism for compact manifolds, $\ker D = \bigoplus_k \mathcal{H}^k(T^2; \mathbb{R})$. The dimension of $\mathcal{H}^k$ is the k -th Betti number, $(b_0, b_1, b_2) = (1, 2, 1)$.

(iii) With $\gamma = (-1)^F$ acting as $+1$ on $\Omega^0 \oplus \Omega^2$, -1 on Ω^1 , we have $\{\gamma, D\} = 0$. If $D\psi = \lambda\psi$, then $D(\gamma\psi) = -\lambda(\gamma\psi)$. The nonzero spectrum is $\pm$ -symmetric. $\square$

Corollary 6.2. *The Hodge–Dirac evades the Nielsen–Ninomiya no-go theorem. Zero modes are topological (harmonic forms), not momentum-space doublers.*

7 The sector-triplet correspondence

Theorem 7.1. *The nine A_1 copies in the $\lambda = 4$ sector split as*

$$9A_1 = 3A_1(\Omega^0) \oplus 3A_1(\Omega^1) \oplus 3A_1(\Omega^2). \quad (29)$$

Proof. Immediate from Theorem 3.4 and its corollaries. Each sector contributes exactly three A_1 copies. Since the sector decomposition is orthogonal and D_4 -invariant, the A_1 isotypic of the direct sum is the direct sum of the sector A_1 isotypics. $\square$

Corollary 7.2. *The A_1 multiplicity space of the full $\lambda = 4$ sector is 9-dimensional, decomposing into three orthogonal 3-dimensional subspaces, one per Hodge sector.*

8 The hypercharge space

8.1 Sector operators

Definition 8.1. *The three sector operators on the $\lambda = 4$ sector are*

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2}, \quad (30)$$

$$\mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad (31)$$

$$\mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (32)$$

Restricted to the three sectors $(\Omega^0, \Omega^1, \Omega^2)$, the sector values are

$$\mathbb{I}|_{\text{sectors}} = (1, 1, 1), \quad \mathcal{F}|_{\text{sectors}} = (1, -1, 1), \quad \mathcal{O}|_{\text{sectors}} = (1, 0, -1). \quad (33)$$

8.2 The 15-dimensional hypercharge space

Definition 8.2. A hypercharge functional on the $\lambda = 4$ sector is a D_4 -invariant, sector-block-diagonal, multiplicity-scalar operator on the 56-dimensional space.

Theorem 8.3. The space of hypercharge functionals is 15-dimensional.

Proof. A hypercharge functional acts as a scalar on each isotypic block (ρ, Ω^k) with $\rho \in \{A_1, A_2, B_1, B_2, E\}$ and $k \in \{\Omega^0, \Omega^1, \Omega^2\}$. There are $5 \times 3 = 15$ such blocks.

For each fixed ρ , the restrictions of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ to the three sectors give three vectors in $\mathbb{R}^3$: $(1, 1, 1)$, $(1, -1, 1)$, $(1, 0, -1)$. The matrix with these rows has determinant

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix} = 1(1 - 0) - 1(-1 - 1) + 1(0 + 1) = 1 + 2 + 1 = 4 \neq 0. \quad (34)$$

So the three operators are linearly independent and span $\mathbb{R}^3$, equal to $\text{span}\{P_\rho^{(0)}, P_\rho^{(1)}, P_\rho^{(2)}\}$. Over the five irreps, the span is $5 \times 3 = 15$ -dimensional. $\square$

8.3 The hypercharge coefficients under Input 1

Under Input 1 (Standard Model content), the three color-triplet multiplets have hypercharges $-2/3, +1/6, +1/3$. These are assigned to the three Hodge sectors as follows.

Assignment of Ω^1 . The B_1 multiplicities across the three sectors are 2, 3, 2. The sector Ω^1 is uniquely distinguished by having B_1 multiplicity 3, matching the two weak-doublet multiplets Q_L and L_L . The color-triplet weak doublet Q_L is the natural occupant of Ω^1 .

Assignment of Ω^0 and Ω^2 . The sectors Ω^0 and Ω^2 have identical D_4 decompositions (they are Hodge-dual to each other). The assignment of u_R^c to one and d_R^c to the other is a labeling convention, fixed by requiring $Y(u_R^c) < Y(d_R^c)$ in the standard ordering of right-handed quark hypercharges. Theorem 9.1 below shows that this labeling is a symmetry of the framework: swapping $\Omega^0 \leftrightarrow \Omega^2$ swaps $u_R^c \leftrightarrow d_R^c$ and leaves all physical predictions invariant.

Theorem 8.4. With $Y(u_R^c) = -2/3$ on Ω^0 , $Y(Q_L) = +1/6$ on Ω^1 , $Y(d_R^c) = +1/3$ on Ω^2 , the unique coefficients $(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) \in \mathbb{R}^3$ for the A_1 block are

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -\frac{1}{6}, -\frac{1}{2}). \quad (35)$$

Proof. The hypercharge functional on the A_1 isotypic is

$$Y_{A_1} = \alpha_{A_1} \mathbb{I} + \beta_{A_1} \mathcal{F} + \gamma_{A_1} \mathcal{O}. \quad (36)$$

Restricted to the three sectors:

$$\Omega^0: \quad \alpha + \beta + \gamma = -\frac{2}{3}, \quad (37)$$

$$\Omega^1: \quad \alpha - \beta = \frac{1}{6}, \quad (38)$$

$$\Omega^2: \quad \alpha + \beta - \gamma = \frac{1}{3}. \quad (39)$$

From the second equation: $\alpha = \beta + 1/6$. Adding the first and third: $2\alpha + 2\beta = -1/3$, so $\alpha + \beta = -1/6$. Substituting: $2\beta + 1/6 = -1/6$, so $\beta = -1/6$, $\alpha = 0$. From the first: $\gamma = -2/3 - \alpha - \beta = -1/2$. $\square$

Theorem 8.5. *With $Y(e_R^c) = +1$ on Ω^0 , $Y(L_L) = -1/2$ on Ω^1 , $Y(\nu_R^c) = 0$ on Ω^2 , the unique coefficients for the B_1 block are*

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +\frac{1}{2}, +\frac{1}{2}). \quad (40)$$

Proof. The system is

$$\alpha + \beta + \gamma = 1, \quad \alpha - \beta = -\frac{1}{2}, \quad \alpha + \beta - \gamma = 0.$$

From the second: $\alpha = \beta - 1/2$. Adding first and third: $2\alpha + 2\beta = 1$, so $\alpha + \beta = 1/2$. Substituting: $2\beta - 1/2 = 1/2$, so $\beta = 1/2$, $\alpha = 0$, $\gamma = 1/2$. $\square$

9 Orientation-flip symmetry

The sectors Ω^0 and Ω^2 have identical D_4 decompositions (Theorem 3.4 and Corollary). They are related by Hodge duality, which maps k -forms to $(2 - k)$ -forms and therefore swaps $\Omega^0 \leftrightarrow \Omega^2$ while fixing Ω^1 . This is a genuine symmetry of the Hodge complex.

Theorem 9.1 (Orientation-flip symmetry). *Let $Y_{A_1} = \alpha\mathbb{I} + \beta\mathcal{F} + \gamma\mathcal{O}$ be the hypercharge functional on the A_1 isotypic of $V_{\lambda=4}$, with coefficients (α, β, γ) . Under the sign flip $\mathcal{O} \mapsto -\mathcal{O}$, the coefficients transform as $(\alpha, \beta, \gamma) \mapsto (\alpha, \beta, -\gamma)$. The physical hypercharge multiset on the sector-triplets is invariant:*

$$\{Y_{A_1}(\Omega^0), Y_{A_1}(\Omega^1), Y_{A_1}(\Omega^2)\} = \{Y'_{A_1}(\Omega^0), Y'_{A_1}(\Omega^1), Y'_{A_1}(\Omega^2)\}. \quad (41)$$

Equivalently, the flip exchanges the labels $u_R^c \leftrightarrow d_R^c$ while leaving all physical predictions invariant.

Proof. The sector values of $\mathcal{O}$ are $(+1, 0, -1)$ on $(\Omega^0, \Omega^1, \Omega^2)$. Under the flip, they become $(-1, 0, +1)$, exchanging Ω^0 and Ω^2 .

The A_1 coefficients from Theorem 8.4 are $(\alpha, \beta, \gamma) = (0, -1/6, -1/2)$. Under the flip, $\gamma \mapsto +1/2$, and the sector values become

$$Y'_{A_1}(\Omega^0) = \alpha + \beta + \gamma' = 0 - \frac{1}{6} + \frac{1}{2} = +\frac{1}{3}, \quad (42)$$

$$Y'_{A_1}(\Omega^1) = \alpha - \beta = +\frac{1}{6}, \quad (43)$$

$$Y'_{A_1}(\Omega^2) = \alpha + \beta - \gamma' = 0 - \frac{1}{6} - \frac{1}{2} = -\frac{2}{3}. \quad (44)$$

Comparing with the unflipped values $(-2/3, +1/6, +1/3)$, the multiset $\{-2/3, +1/6, +1/3\}$ is preserved. The flip exchanges the roles of Ω^0 and Ω^2 , i.e., $u_R^c \leftrightarrow d_R^c$.

The physical predictions of the framework — the three hypercharge values (as a multiset), the anomaly cancellation conditions, and the CKM/PMNS structure (which depends only on the multiset of hypercharges, not their sector-by-sector labels) — are invariant under this exchange. $\square$

Corollary 9.2 (Labeling convention). *The sign of $\mathcal{O}$ is a labeling convention, fixed by the requirement $Y(u_R^c) < Y(d_R^c)$. Flipping the convention produces an equivalent formulation with no physical consequence.*

Remark 9.3. *The symmetry is not merely a mathematical curiosity. It is the reflection of the fact that the framework does not privilege Ω^0 over Ω^2 . The two sectors are Hodge-dual, carry identical D_4 representations, and are distinguished only by the sign convention for the orientation operator $\mathcal{O}$. The framework derives the space of hypercharge functionals and the $\mathbb{Z}_2$ symmetry that acts within it; the choice of which sector is called u_R^c and which is d_R^c is a labeling convention, not a physical input.*

10 Anomaly cancellation as a consistency condition

The following is not a theorem about the framework deriving anomaly cancellation from first principles. It is a consistency check: given Input 1 (the Standard Model content), the hypercharge configuration selected by Theorems 8.4 and 8.5 satisfies the four anomaly cancellation conditions exactly. This verifies that Input 1 is compatible with the framework's algebraic structure.

Proposition 10.1 (Anomaly consistency). *The Standard Model content per generation, equation (2), with three generations, satisfies the four anomaly conditions*

$$\text{Tr}(Y) = 0, \quad \text{Tr}(Y^3) = 0, \quad \text{SU}(2)^2\text{U}(1) = 0, \quad \text{SU}(3)^2\text{U}(1) = 0 \quad (45)$$

exactly.

Proof. Using Dynkin indices $T(2) = T(3) = 1/2$.

$\text{Tr}(Y)$:

$$\text{Tr}(Y) = 3 \cdot \left[6 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3} \right) + 3 \cdot \frac{1}{3} + 2 \cdot \left(-\frac{1}{2} \right) + 1 \cdot 1 + 1 \cdot 0 \right] = 3 \cdot 0 = 0.$$

$\text{Tr}(Y^3)$:

$$\begin{aligned} \text{Tr}(Y^3) &= 3 \cdot \left[6 \cdot \left(\frac{1}{6} \right)^3 + 3 \cdot \left(-\frac{2}{3} \right)^3 + 3 \cdot \left(\frac{1}{3} \right)^3 + 2 \cdot \left(-\frac{1}{2} \right)^3 + 1 \cdot 1^3 + 1 \cdot 0^3 \right] \\ &= 3 \cdot \left[\frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1 \right] \\ &= 3 \cdot \left[\frac{1}{36} - \frac{32}{36} + \frac{4}{36} - \frac{9}{36} + \frac{36}{36} \right] \\ &= 3 \cdot \frac{0}{36} = 0. \end{aligned}$$

$\text{SU}(2)^2\text{U}(1)$:

$$\text{SU}(2)^2\text{U}(1) = 3 \cdot \frac{1}{2} \cdot \left[3 \cdot \frac{1}{6} \cdot 2 + 1 \cdot \left(-\frac{1}{2} \right) \cdot 2 \right] = 3 \cdot \frac{1}{2} \cdot [1 - 1] = 0.$$

$\text{SU}(3)^2\text{U}(1)$:

$$\text{SU}(3)^2\text{U}(1) = 3 \cdot \frac{1}{2} \cdot \left[\frac{1}{6} \cdot 2 + \left(-\frac{2}{3} \right) \cdot 1 + \frac{1}{3} \cdot 1 \right] = 3 \cdot \frac{1}{2} \cdot \left[\frac{2}{6} - \frac{4}{6} + \frac{2}{6} \right] = 0.$$

□

Corollary 10.2. *The cubic gauge anomalies $\text{SU}(3)^3$, $\text{SU}(2)^3$, and the mixed $\text{SU}(3)$ - $\text{SU}(2)$ - $\text{U}(1)$ vanish by the same computation.*

Corollary 10.3 (Two-loop automatic). *By the Adler–Bardeen theorem, the ABJ anomaly is one-loop exact; vanishing one-loop conditions imply vanishing at all loops.*

Remark 10.4. *The anomaly consistency check is a verification that Input 1 is compatible with the framework's algebraic structure, not a derivation of anomaly cancellation from first principles. The same check is standard practice in any framework that takes the Standard Model content as input, including the Standard Model itself.*

11 Summary of derived results and inputs

11.1 Theorems (derived)

- T1: $D^2 = \Delta$ (Theorem 2.3).
- T2: $\lambda = 4$ multiplicity 14 (Theorem 3.2).
- T3: D_4 decompositions at $\lambda = 4$ (Theorem 3.4 and Corollaries).
- T4: Commutant dimensions 27, 100, 400 (Corollaries of Theorem 4.2).
- T5: Gauge algebra embedding (Theorem 5.5).
- T6: Nielsen–Ninomiya evasion (Theorem 6.1).
- T7: Sector-triplet correspondence (Theorem 7.1).
- T8: 15-dimensional hypercharge space (Theorem 8.3).
- T9: Orientation-flip symmetry (Theorem 9.1).

11.2 Inputs

- Input 1: The Standard Model per-generation content, equation (2).
- Input 2: The physical identification of the A_1 isotypic with color and the B_1 isotypic with weak isospin.

11.3 Consistency conditions

- Anomaly cancellation: Proposition 10.1, a verification that Input 1 is compatible with the framework’s structure.

11.4 Not derived

- The specific hypercharge values $-2/3, +1/6, +1/3, -1/2, +1, 0$ are not derived from the framework. They are the content of Input 1. The framework derives the space in which they live and the symmetry constraints they satisfy.
- The physical identification of the algebraic structures with the Standard Model gauge groups is Input 2. This is the same identification every framework must make.

The paper derives the algebraic structure that any physically realized gauge theory with the Standard Model content must satisfy, and proves that the Standard Model content is consistent with this structure.

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